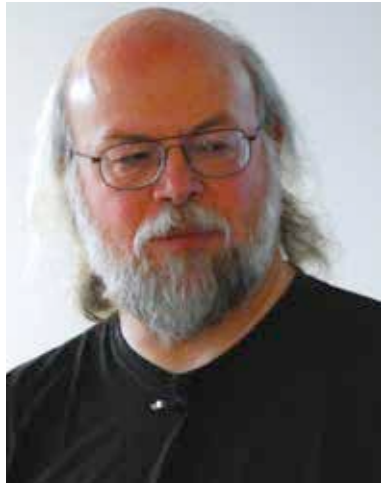


Java is used to develop enterprise-wide applications, web applications, games and now mobile applications.

In the mobile age of today Java is more relevant than ever. Android programming is largely done using Java. In fact the whole Oracle v/s Google lawsuit was because Oracle claimed Google infringed on their Java APIs to develop the Android operating system and APIs.



James Gosling

Java Editions

There are four editions of Java that target different application environments. The Java APIs are segmented to belong to one of the editions.

- ◆ Java Card for Smart Cards
- ◆ Java Platform, Micro Edition (ME) for environments with limited resources such as mobiles.
- ◆ Java Platform, Standard Edition (SE) for workstation environments
- ◆ Java Platform, Enterprise Edition (EE) for large distributed enterprises or Internet environments.

There are also a number of lightweight languages built on top of Java that can use the JVM eg Groovy (a dynamic language with features of Python, Perl, Smalltalk and Ruby), Clojure, Jython (a Python interpreter), JRuby (a Ruby interpreter). Each of these were built for their own dedicated purpose.

Java and Android

The Android operating system is built on the Linux kernel which is written in C. Android applications however are written in Java. Android does not use the Java Virtual Machine or JVM. It has its own Virtual Machine called the Dalvik Virtual Machine which does all the byte code conversions. The Java code is compiled into proprietary byte code and run on the Dalvik Virtual Machine. The Dalvik VM has a register based architecture unlike the Java VM which has uses a stack based architecture.